



Figure 4: **Efficiency Analysis.** Evolution of private state size (in tokens) over dialogue turns on a logarithmic scale. Shaded regions indicate the 95% confidence interval. **Private CoT** (Grey) requires approximately 10x more context than explicit memory methods.

confirming plausible candidates, suggesting it is more "comfortable" fabricating the existence of a secret state to satisfy the immediate verification prompt. Finally, Qwen3 32B in the Diagnosis Simulator displays a strong tendency toward Leakage, likely driven by safety alignment training that prioritizes transparency and helpfulness over the task's requirement that the secret remain unresolved.

RQ3: What is the token overhead of maintaining private state? Figure 4 compares the token cost of maintaining hidden information across methods. We define "Private State Size" as the total count of tokens injected into the context window at each turn, excluding the public dialogue history. For Private CoT, this includes the cumulative history of all reasoning traces; for retrieval baselines (Mem0, A-Mem, LightMem, and MemoryOS), it is the size of the retrieved context; and for our Private Working Memory agents, it is the length of the maintained text block.

As expected, Private CoT has a state size approximately one order of magnitude larger than all other methods. By retaining every intermediate reasoning step, the context expands rapidly, making it unscalable for long-horizon interactions. In contrast, our Private Working Memory agents maintain a compact footprint (typically under 100 tokens), effectively compressing the necessary state into a concise format that remains constant or grows slowly, comparable to standard retrieval baselines.

6 DISCUSSION

The Generative-Retention Gap. Our results highlight a critical distinction between established memory-augmented architectures and the requirements of Private State Interactive Tasks. Current memory systems (e.g., Mem0, A-Mem) are primarily designed for the *passive accumulation* of context; they excel at efficiently summarizing, annotating, and recalling information explicitly present in the public conversation history to maintain long-horizon coherence. However, they lack the mechanism to store self-generated information, such as the internal reasoning traces required to form a private plan or secret. In contrast, solving PSITs requires a **Generative-Retention Loop**: the agent must not only recall existing data but generate new information (e.g., a secret word, a plan, or a reasoning step) and immediately retain it for future turns. While modern reasoning models possess the capability to generate these intermediate reflections, they lack the persistence layer to store them. Standard memory baselines fail because they treat memory as a log of the interaction, whereas autonomous agents require memory as a container for cognition. We designed tasks like Hangman specifically to expose this gap, demonstrating that the ability to generate a thought is futile without the architectural capacity to persist it.

Hidden State vs. Helpfulness. Our failure mode analysis further reveals a tension between architectural capability and safety alignment. Without a private memory channel, models are forced into a trade-off: they either hallucinate a state to continue the interaction or deny the state to remain truthful. Notably, models with strong safety alignment (e.g., Qwen3) frequently default to Leakage, revealing the secret to the user. This suggests that without an explicit private workspace, the "helpfulness" objective of RLHF overrides the constraint of the game. Private Working Memory resolves this tension by providing a designated space where the model can be "honest" about its state without violating the requirement that the secret remain unresolved in the public interface.

7 CONCLUSION

As LLMs evolve from text completers to autonomous agents, the standard chat interface becomes inadequate for tasks requiring persistent hidden state. We formalize this setting as Private State Interactive Tasks (PSITs) and show theoretically that agents conditioned only on public history cannot simultaneously keep the secret unresolved in the public transcript and remain consistent with a fixed hidden state. Empirically, we find that standard LLMs and retrieval-based baselines fail these tasks, exhibiting either state loss or information leakage. Our results highlight Private Working Memory as a necessary architectural component for PSITs. By explicitly maintaining private state, agents can separate internal reasoning from public actions and recover consistency with a fixed hidden state. Future research should explore additional interactive tasks where private working memory can improve an agent’s ability to manage complex internal states.

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